Yiwei Wang

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Education

Technical University of Munich (TUM)
Oct 2026 – Present

M.Sc. in Mechatronics, Robotics and Biomechanical Engineering
Munich, Germany

University of Nottingham Ningbo China
Sep 2020 – Nov 2025

B.Eng. (Hons) in Aerospace Engineering
Ningbo, China

- **Degree:** Upper Second Class Honours (2:1), awarded by the University of Nottingham, UK; **WES GPA 3.74/4.0**; **top 15%** of the cohort.
- **Key coursework:** Control of Aerospace Systems (87), Aerospace Statics & Dynamics (80), Computer Modelling Techniques (76), Foundation Calculus (89), Foundation Algebra (86).

Highlights

- **2nd place among 2,000+ international teams** (tied with ETH Zürich) as team leader, ASME K-16 / IEEE EPS Heat Sink Design Competition 2024.
- **Embodied-AI industry experience** at CAICT: contributed to an industry robot-capability standard and tested robots in live 5G networks.
- **Former professional Go (Weiqi) player** with national-level competition results in China.

Research Interests

I am interested in **learning dexterous manipulation from human videos**, a promising way around the data bottleneck of robot teleoperation. At CAICT, I helped draft an industry standard on evaluating factory robots and tested material-handling robots in live deployments, where I saw how **brittle today's systems are outside the conditions they were tuned for** and how hard it is to compare them fairly. My hands-on work on multi-axis kinematics (5-axis CNC with RTCP) also draws me to a key open problem: **transferring human hand motion to robot hands with different embodiments**.

Industry Experience

China Academy of Information and Communications Technology (CAICT)
May 2026 – Sep 2026

Research Intern, Institute of Technology and Standards
Beijing, China

Advisors: Dr. Ying Huang, Dr. Min Chen

- **Embodied-AI standards:** contributed to the AIIA standard *Capability Requirements and Evaluation Methods for Factory Material-Handling Robots*; prepared materials for the AIIA working-group meeting (Aug 2026).
- **Robot testing:** wrote a technical verification report for a **mobile manipulator** operating in a live 5G network; ran industrial 5G terminal tests on an explosion-proof inspection robot, a sample-delivery robot and a smart meter reader.
- **Research reports:** co-authored a report on 5G-Advanced in industrial scenarios (Chinese and English editions); surveyed embodied-AI technologies and products across companies and application scenarios.

ZYT Petroleum Equipment Co., Ltd.
Jan 2026 – May 2026

Mechanical Engineering Intern
Puyang, China

- Machined and assembled key modules of mobile drilling rigs (derricks, winches, power and transmission systems); modified component designs to customer requirements.
- Assembled and debugged **hydraulic, pneumatic and electronic control units**; took part in inspection, functional testing and quality control.

Industry Collaboration Project (diamond-cutting application)
Sep 2022 – Jan 2023

Student Engineer, 5-Axis CNC Machine Tool
Ningbo, China

- Designed and commissioned a **5-axis CNC machine** for industrial diamond cutting; scoped requirements, budget and trade-offs directly with the client.
- Programmed tool-path generation in **C#** and implemented **real-time multi-axis interpolation with RTCP**; achieved coordinated motion via **encoder-feedback synchronization** and error analysis.
- Completed mechanical assembly and electrical debugging; integrated limit switches and emergency stop on an Arduino I/O board.

Research Experience

University of Nottingham Ningbo China
Sep 2024 – May 2025

Final-Year Thesis, advised by Dr. Yi Nie
Ningbo, China

- **Research topic: high-throughput powder spreading for Laser Powder Bed Fusion (LPBF)**, bringing high-throughput screening concepts from the pharmaceutical industry into metal additive manufacturing.

- Architected a **16-piston, point-to-point powder dosing module** that deposits multiple distinct alloy formulations in a **single print cycle**, greatly increasing alloy-screening throughput.
- Implemented motion sequencing and synchronization in **C# (.NET)**; validated the mechanical layout in SolidWorks and 3DExperience.
- Outcomes: **granted patent** [P1]; presented at **CSIAED 2024**.

ASME K-16 / IEEE EPS International Heat Sink Design Competition
Dec 2023 – May 2024

Team Leader
Colorado, USA

- Led a four-person team to **2nd place (tied with ETH Zürich) among 2,000+ international teams**, many of them master's and PhD teams.
- Designed a lightweight heat sink combining **biomimetic geometry** with parametric design exploration; modeled in SolidWorks and simulated in **ANSYS Fluent**.
- Extended the **surrogate-modeling** method to ducted-fan blade design, resulting in a first-author conference paper [C1] (under review).

University of Nottingham Ningbo China
Jan 2023 – May 2023

Research Project: Control & Stability of a VTOL Drone, advised by Dr. Salman Ijaz
Ningbo, China

- Derived forward and inverse kinematics of a VTOL drone and implemented attitude and path controllers in **MATLAB/Simulink**.
- Validated closed-loop performance with **hardware-in-the-loop simulation** in FlightGear; tuned PID gains and trajectory tracking under disturbances.

Personal AI Project

Encoding Investment Expertise as LLM Agent Skills
Ongoing

Personal project

- Distilling an experienced investor's **A-share stock-selection methodology** into a structured **LLM skill system** (SKILL.md) with rolling judgment ledgers and calibration logs, so an AI agent can apply and self-check the method.
- Built a parallel **Python data pipeline** for Chinese A-share market data using **AKShare**.

Publications & Patents

[C1] Surrogate-based Multifidelity Robust Optimization for Ducted Fan Blade Aerodynamic Design
Yiwei Wang, et al.
*IEEE International Conference on Mechanical, Electrical and Aerospace Engineering (MEAE), **under review***

[P1] High-Throughput Powder Spreading System for Laser Powder Bed Fusion of Multi-Compositional Alloys
Yiwei Wang, Yi Nie
*Patent, **granted***

Skills

Programming	Python, C++, C#, MATLAB, Arduino (C/C++)
AI Tools	LLM agents (Claude, Codex), agent skill design, Python data pipelines
Simulation	MATLAB/Simulink (UAV Toolbox), ANSYS (Fluent, CFX), FlightGear, COMSOL (basic)
CAD / CAE	SolidWorks, 3DExperience, CATIA, FEA, CFD
Methods	Control (PID, RTCP, multi-axis interpolation), kinematics, surrogate modeling, additive manufacturing (LPBF)
Languages	Chinese (native), English (professional)

Leadership & Activities

- **President, UNNC Chess Club** (2022–2023): organized weekly sessions and a campus tournament; won the university's **Dream Scholarship** for the club with a program proposal.
- Long-time basketball player and fitness enthusiast.